Ankush Kapoor

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Education

Vidyalankar Institute of Technology — **Mumbai, India**
Bachelor of Technology in Computer Engineering — *Sept 2024 – Present*
- CGPA: **10.0** (Semesters 1-4) — Expected Graduation: **2028**

Experience

Open World Holidays Framework — ***Remote*** | **GSoC Project Page**
Google Summer of Code 2026 Contributor (Python Software Foundation) — *May 2026 – Oct 2026*
- Implemented verified holiday coverage for **India** at **national** level and across all **28 states + 8 UTs**, cross-referenced against official government notices and calendars.
- Built a **solar and lunisolar calendar generator** for Hindu festivals including **Diwali, Holi, and Makar Sankranti**, using astronomical and mathematical calculations (tithi, paksha, sunrise) to cover approximately **60 holidays**.
- Added localized holiday names across **10 Indian languages** for regional users.

Open Source Contributor (GWoC Rank 7/2300+) & Sponsored Contributor — *Jan 2025 – Jun 2025*
- Achieved **Rank 7 out of 2300+ contributors** in Google Winter of Code (GWoC) '25; top-tier performance led to a **sponsored task** to implement **Mongolian holidays**.
- Implemented a **lunar calendar** for **Mongolian holidays** and extended existing holiday-generation systems used for **India** and **Nepal**.
- Added Hindi and Mongolian translations, improving native-language accessibility for regional users.

Projects

BaseKernel (Custom 32 bit Kernel) | *C, NASM, QEMU* — **GitHub**
- Built a **32-bit protected mode** kernel from scratch, **bootstrapped** from **real mode**.
- Implemented **bootloader**, **GDT**, **IDT**, **ISR**, and **IRQ** handling for low-level system control.
- Added **memory paging**, **screen output** along with a **Round Robin scheduler**.
- Developed entirely without standard libraries to achieve bare-metal execution.

Transformer From Scratch (Neural Machine Translation) | *Python, PyTorch* — **GitHub**
- Developed a complete **Transformer architecture** from scratch using **PyTorch**, manually building **multi-head attention**, **encoder-decoder blocks**, and **positional encoding**.
- Trained a bilingual **NMT model** on **OPUS Books** using **greedy and beam search** decoding, implementing **learning-rate warmup** and **label smoothing** for optimized convergence.
- Evaluated performance using **SacreBLEU, WER, and CER** metrics, utilizing **TensorBoard** for experiment tracking and checkpointing logic.

Alloc (Custom Memory Allocator in C) | *C, NASM* — **GitHub**
- Created a 32-bit allocator mimicking **malloc()** and **free()** over a 1 GB virtual heap.
- Implemented **alloc(bytes)** to allocate memory in words by taking size in bytes as input.
- Added **destroy()** for secure deallocation with memory zeroing to prevent data leakage, ensuring data privacy.
- Built packed headers, manual tracking, macros for KB/MB/GB allocation, and a **show()** debugger.

Technical Skills

Languages: C, Python, Assembly, Java — **Database**: MySQL
Core Areas: Deep Learning, Machine Learning, Low-Level Systems Programming
Frameworks & Tools: PyTorch, Scikit-learn, XGBoost, Git, GitHub, GCC, NASM, QEMU
OS: Linux (Ubuntu, WSL), Windows

Certifications & Achievements

- **3rd Place** in Cognitive Clash (AI-ML Showdown) Hackathon - built an **AI Virtual Try-On** System — Oct 2026
- Machine Learning with Python – L&T Skill Trainers Academy, Mumbai (Residential Course) — Jun 2026

Extracurricular Activity

Our Tech Community (OTC) | ourtech.community — **Jul 2025 – Present**
Co-Organiser
- Co-organize and grow a vibrant, inclusive, and open-for-all tech community of currently **600+ members**.
- Plan, organize, and execute OTC's **flagship programs**, including weekly catchups and **technical** talks.
- Oversee community operations, including **technical setup and maintenance** for events, ensuring smooth execution and **comprehensive documentation**.

Rotary Divyang Center | rotarydivyangcenter.org — **Feb 2026 – Apr 2026**
Social Service Intern
- Supported operations at a **prosthetics and orthotics center** providing **artificial limbs**, assisting patients with administration and coordination.
- Built and deployed the center's website to enhance **outreach and accessibility**.